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An example of a continuous LTI system response solution

[EX.] The differential equation describing a continuous LTI system is

$$y''(t)+7y'(t)+12y(t)=x(t),\quad t\geq 0$$

Excitation signal: $x(t)=u(t)$, Initial state: $y(0^-)=1$, $y'(0^-)=2$

Try:

(1) The system has zero input response $y_{zi}(t)$

(2) The impulse response of the system $h(t)$;

(3) The system responds with zero state $y_{zs}(t)$;

(4) full response of the system $y(t)$;

(5) Determine whether the system is stable.

An example of a continuous LTI system response solution

[EX.] The differential equations that describe a continuous LTI system a known to be

$$y''(t)+7y'(t)+12y(t)=x(t),\quad t\geq 0$$

Excitation signal: $x(t)=u(t)$, Initial state: $y(0^-)=1$, $y'(0^-)=2$

Solution: (1) The system has zero input response $y_{zi}(t)$

characteristic equation: $s^2+7s+12=0$

characteristic root: $s_1=-3, s_2=-4$ $\Rightarrow$ $y_{zi}(t)=K_1e^{-3t}+K_2e^{-4t},\quad t\geq 0^-$

Substitute the initial state: $y(0^-)=K_1+K_2=1$ $\Rightarrow$ $K_1=6,\ K_2=-5$

$$y'(0^-)=-3K_1-4K_2=2$$

$$y_{zi}(t)=6e^{-3t}-5e^{-4t}, t\geq 0^-$$

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Excitation signal $x(t)=u(t)$, Initial state: $y(0^-)=1$, $y'(0^-)=2$

(2) Impulse response of the system $h(t)$ $h''(t)+7h'(t)+12h(t)=\delta(t)$

Using the impulse balance method, let h(t) take the form of

$$h(t)=(Ae^{-3t}+Be^{-4t})u(t)$$

substitute $h''(t)+7h'(t)+12h(t)=\delta(t)$,

$4A+3B=1$ get the undetermined coefficient: $A=1$, $B=-1$。

$A+B=0$ The available impulse response is $h(t)=(e^{-3t}-e^{-4t})u(t)$

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Excitation signal: $x(t)=u(t)$, Initial state: $y(0^-)=1$, $y'(0^-)=2$

(3) The system has zero input response

The zero-state response is equal to the convolution of the system input signal and the impulse response.

$$y_{zs}(t)=x(t)*h(t)=u(t)*(\mathrm{e}^{-3t}-\mathrm{e}^{-4t})u(t)$$

$$=(\frac{1}{12}-\frac{1}{3}\mathrm{e}^{-3t}+\frac{1}{4}\mathrm{e}^{-4t})u(t)$$

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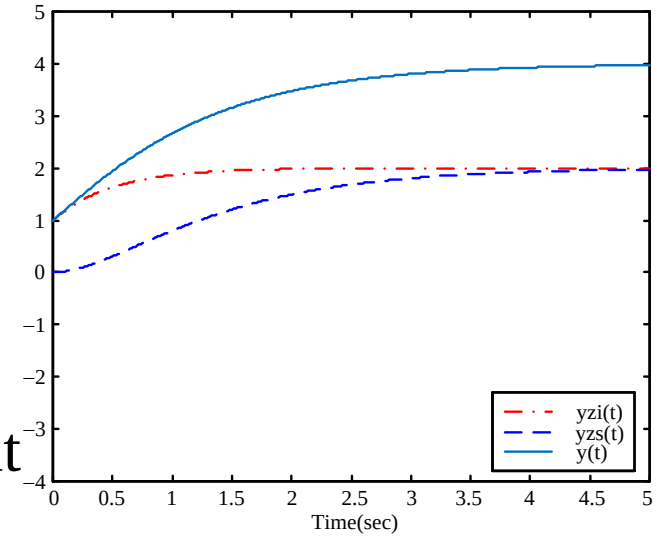
Excitation signal: $x(t)=u(t)$, Initial state: $y(0^-)=1$, $y'(0^-)=2$

(4) The complete response of the system is

$$y(t)=y_{zi}(t)+y_{zs}(t)$$

$$=\frac{1}{12}+\frac{17}{3}e^{-3t}-\frac{19}{4}e^{-4t},\quad t>0$$

$y_{zi}(t)$, $y_{zs}(t)$ and $y(t)$ are shown on the right



An example of a continuous LTI system response solution

[EX.] The differential equations that describe a continuous LTI system a known to be

$$y''(t)+7y'(t)+12y(t)=x(t),\quad t\geq 0$$

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(4) The complete response of the system is

$$y(t)=y_{zi}(t)+y_{zs}(t)=\frac{1}{12}+\frac{17}{3}e^{-3t}-\frac{19}{4}e^{-4t},\quad t>0$$

The complete response can also be decomposed into：

- Intrinsic responses $y_h(t)$ and forced responses $y_p(t)$
- Transient response $y_t(t)$ and Steady-State Response $y_s(t)$

An example of a continuous LTI system response solution

Intrinsic response $y_h(t)$: The response corresponding to the characteristic root of the differential equation of the system.

Forced response $y_p(t)$: The response corresponding to the input signal of the system.

Transient response $y_t(t)$: A component that grows towards 0 over time.

steady state response $y_s(t)$: A component that does not tend to zero over time.

An example of a continuous LTI system response solution

[EX.] The differential equations that describe a continuous LTI system a known to be $y''(t)+7y'(t)+12y(t)=x(t),\quad t\geq 0$

Excitation signal $x(t)=u(t)$, Initial state $y(0^-)=1$, $y'(0^-)=2$

(4) The complete response of the system is

$$y(t)=y_{zi}(t)+y_{zs}(t)=\frac{1}{12}+\frac{17}{3}e^{-3t}-\frac{19}{4}e^{-4t},\quad t>0$$

$$\begin{cases} y_{固有}(t)=\dfrac{17}{3}e^{-3t}-\dfrac{19}{4}e^{-4t}, & t>0 \\ y_{强迫}(t)=\dfrac{1}{12}, & t>0 \end{cases}$$

$$\begin{cases} y_{暂态}(t)=\dfrac{17}{3}e^{-3t}-\dfrac{19}{4}e^{-4t}, & t>0 \\ y_{稳态}(t)=\dfrac{1}{12}, & t>0 \end{cases}$$

An example of a continuous LTI system response solution

[EX.] The differential equations that describe a continuous LTI system a known to be $y''(t)+7y'(t)+12y(t)=x(t),\quad t\geq 0$

Excitation signal $x(t)=u(t)$, Initial state $y(0^-)=1$, $y'(0^-)=2$

(5) Determine if the system is stable

The impulse response of this continuous-time LTI system is

$$h(t)=\mathrm{e}^{-3t}u(t)-\mathrm{e}^{-4t}u(t)$$

$$\int_{-\infty}^{\infty}|h(\tau)|\mathrm{d}\tau=\int_{0}^{\infty}(\mathrm{e}^{-3t}-\mathrm{e}^{-4t})\mathrm{d}\tau=1/12<\infty$$

The continuous LTI system is a stable system

An example of a continuous LTI system response solution

[EX.] If in the example The excitation signal changes to $x_1(t)=0.5u(t-1)$ Resolve the zero-input response of the system $y_{zi}(t)$, zero state response $y_{zs}(t)$ and complete response $y(t)$.

Solution:

Since the initial state of the system does not change, the zero-input response of the system does not change, that is,

$$y_{zi}(t) = 6e^{-3t} - 5e^{-4t}, \quad t \geq 0^-$$

Excitation signal $x_1(t) = 0.5u(t-1) = 0.5x(t-1)$

Using the linear and time-invariant characteristics of the system,

The zero-state response of the system can be obtained as

$$y_{1zs}(t) = 0.5y_{zs}(t-1) = \left(\frac{1}{24} - \frac{1}{6}e^{-3(t-1)} + \frac{1}{8}e^{-4(t-1)}\right)u(t-1)$$

An example of a continuous LTI system response solution

[EX.] If in the example The excitation signal changes to $x_1(t)=0.5u(t-1)$
Resolve the zero-input response of the system $y_{zi}(t)$, zero state response $y_{zs}(t)$ and complete response $y(t)$.

Zero input response of the system

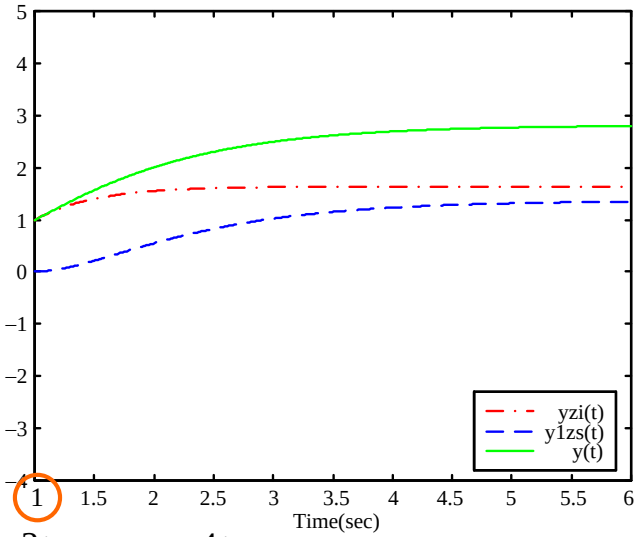
$$y_{zi}(t) = 6e^{-3t} - 5e^{-4t}, \quad t \geq 0^-$$

Zero-state response of the system

$$y_{1zs}(t) = \left(\frac{1}{24} - \frac{1}{6}e^{-3(t-1)} + \frac{1}{8}e^{-4(t-1)}\right)u(t-1)$$

$$y(t) = y_{zi}(t) + y_{1zs}(t)$$

$$= \left(\frac{1}{24} - \frac{1}{6}e^{-3(t-1)} + \frac{1}{8}e^{-4(t-1)}\right)u(t-1) + 6e^{-3t} - 5e^{-4t}, \quad t > 0$$



An example of a continuous LTI system response solution

Summarize：

1. The time domain analysis of the system gives the time domain description of the continuous LTI system.

2. The time domain analysis of the system reveals the interaction mechanism between the signal and the system.

3. The time domain analysis of the system is based on the time domain analysis of continuous signals.

Thanks

Some materials quoted in this course are based on the teaching accumulation of the lecturer for many years. They come from various media and exchanges with colleagues, peers and friends. It is difficult to indicate the source one by one. I hereby explain and express my thanks!